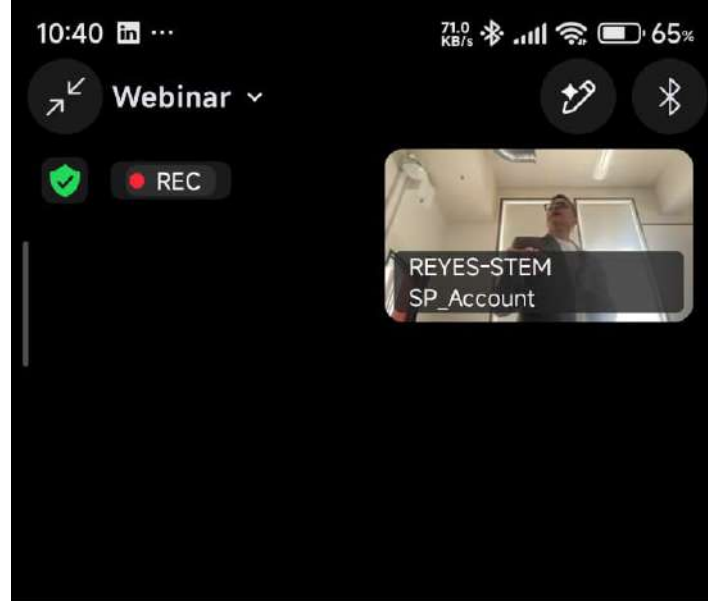
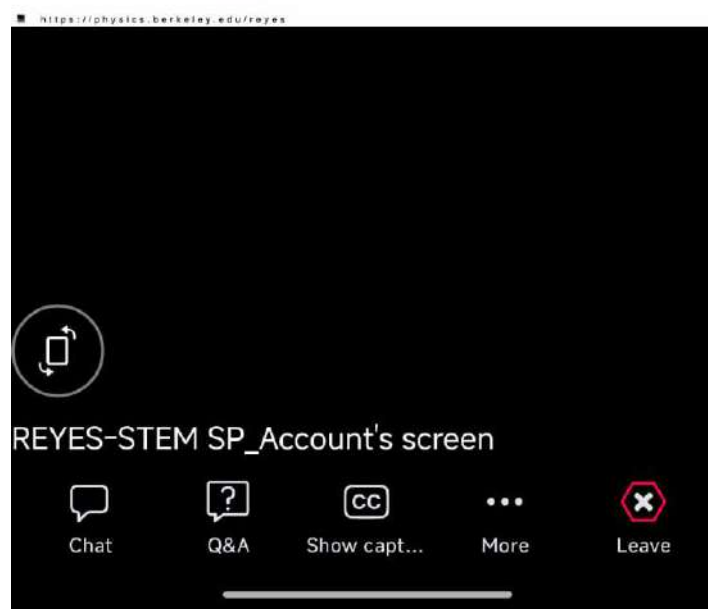


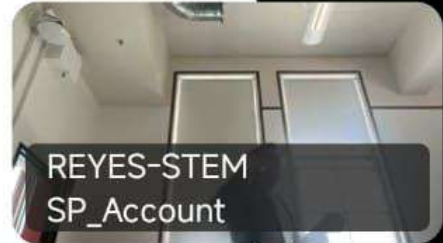


REYES



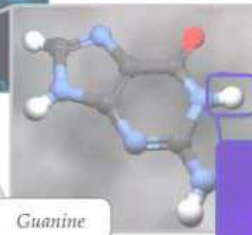
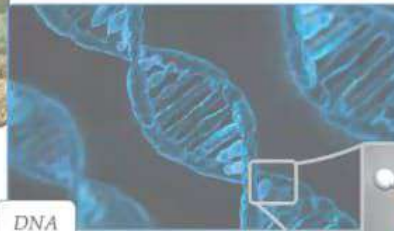
Python4Physics expectations

- Three classes per week: MWF @ 10am (PT),
- Each class will introduce new ideas and exercises,
- Except June 19 and July 3,
- Classes will go until July 10th,
- Videos will be made available for on-line students.

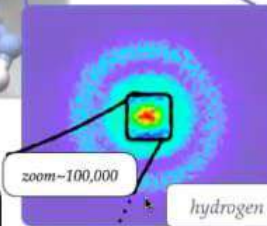




DNA



Guanine

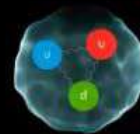


zoom~100,000

hydrogen

the proton:

- the lightest of all nuclei
- carries 99.95% of hydrogen's mass
- made of quarks and gluons



Proton



REYES-STEM
SP_Account

"etched into the cosmos"

zoom~100,000

REYES-STEM
SP_Account

hydrogen

Standard Model

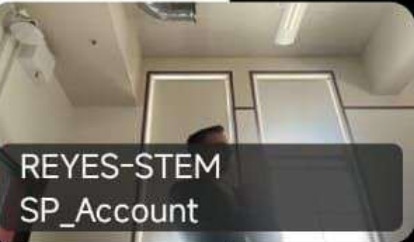
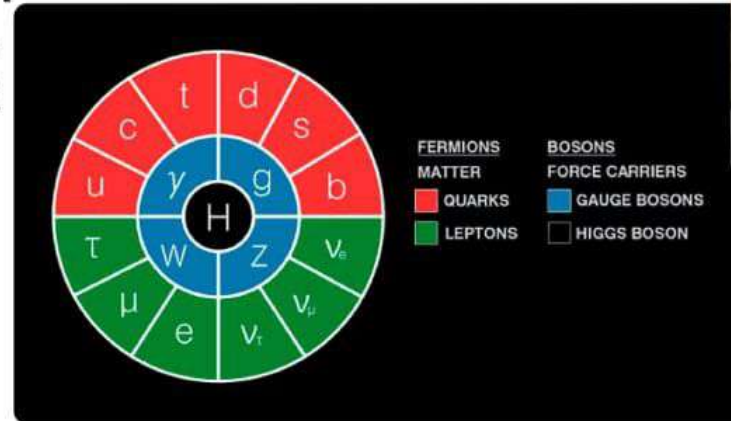
$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + i\bar{\psi}\not{D}\psi + h.c. \\ + \psi_i^\dagger y_{ij} \psi_j \phi + h.c. + |D_\mu \phi|^2 - V(\phi)$$



My research in a nut shell

the Standard Model of Particle Physics

"3/4 of the forces we know of in nature"



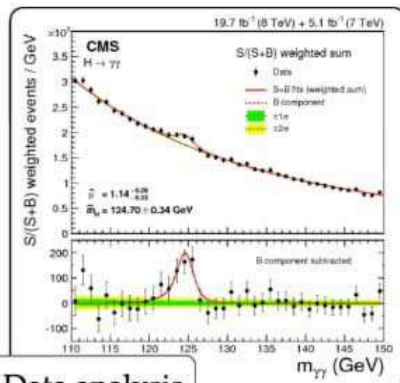
REYES-STEM
SP_Account

RS

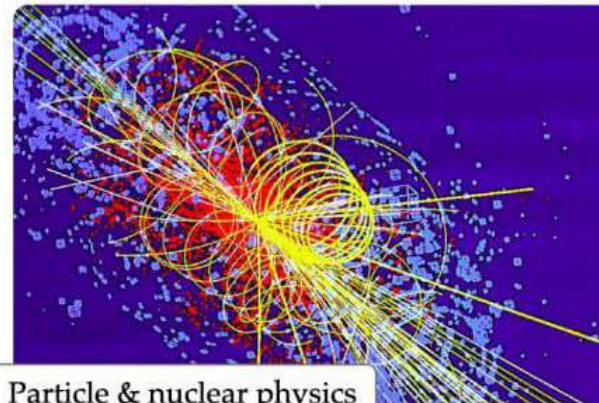
And then the electromagnetic force. These last three are described by the standard model of particle physics. They describe subatomic dynamics and it's a model that in some cases we can predict

REYES-STEM

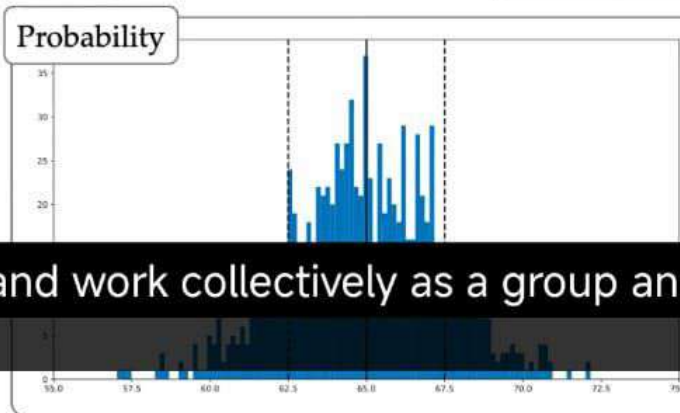
You will learn about...



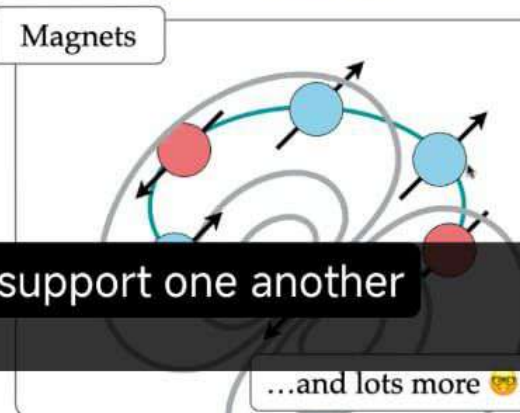
Data analysis



Particle & nuclear physics



Probability



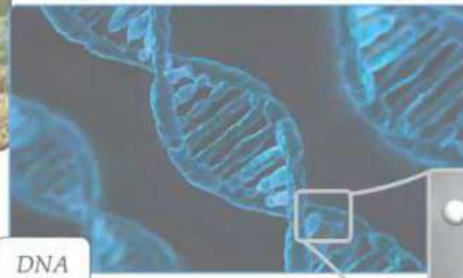
Magnets

RS

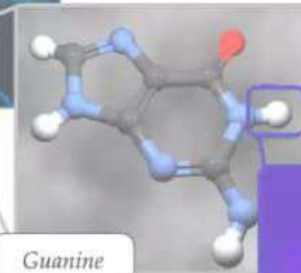
behind and work collectively as a group and support one another

...and lots more 🤖

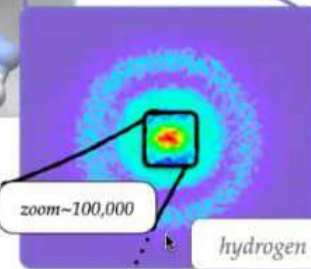




DNA



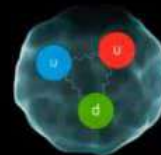
Guanine



hydrogen

the proton:

- 👤 *the lightest of all nuclei*
- 👤 *carries 99.95% of hydrogen's mass*
- 👤 *made of quarks and gluons*



Proton

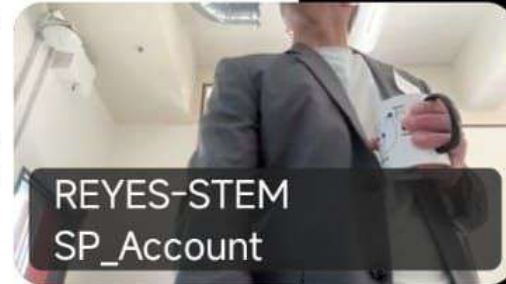


REYES-STEM
SP_Account



“THE UNREASONABLE EFFECTIVENESS OF MATHEMATICS IN THE NATURAL SCIENCES”

Eugene Wigner (1902-1995)



$$\mu = -(1 + a) \frac{eS}{m}$$

✓ unreasonably effective!

$$a[\text{exp}] = 0.00115965218059(13)$$

$$a[\text{th.}] = 0.001159652181643(764)$$



